

InternVideo2: Scaling Foundation Models For Multimodal Video Understanding

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视频分析是人工智能方法的核心部分

数据驱动智能化方法



YouTube | 8M



EPIC
KITCHENS

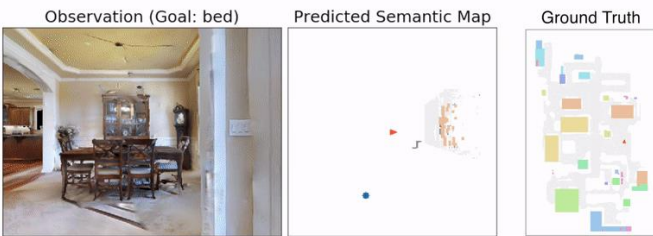
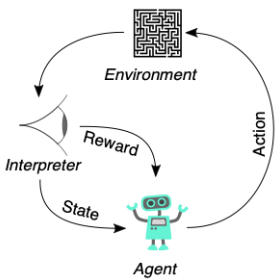
视频是最常见、最丰富数据形态

知识驱动智能化方法



视频是过程式知识的重要表达形式

自主式的智能化方法



视频是感知世界和交互反馈重要载体

视频理解是国家信息和人工智能产业发展的重要需求

人工智能是发展新质生产力的重要引擎，多模态视频理解是当前国际前沿核心之一，也是我国数字产业的技术焦点之一。

2023年我国短视频账号达到**15.5亿**

- 视听业务为主企业**66万**余家
- 人均单日视频应用使用达**187分钟**



2023年我国视频监控摄像头部署
量达**8.64亿**，全球工业摄像头产
值预估达**21亿美元**

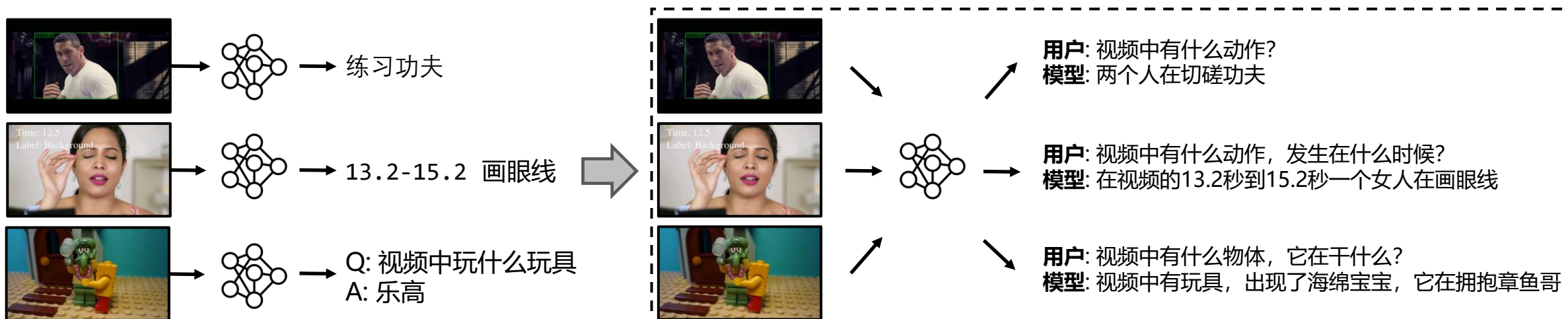


OpenAI旗舰型多模态模型**GPT4o**，
能够实时跨越**视频**、音频和文本进
行推理



核心问题

如何增强视频理解的可迁移性（或者通用性）



海量视频数据及标注

大规模自监督、弱监督学习

适配多种视频下游任务

技术挑战

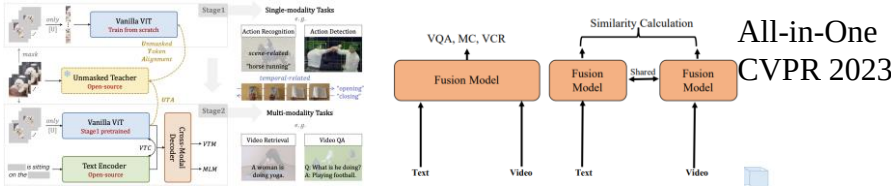
规模化标注效率质量低

视频表征通用性有限

视频任务拓展成本高

相关工作

UMT ICCV2023



VideoMAE NeurIPS2022



2021年前

掩码视频建模和Next token prediction等方法在发展

2022至今

初步研究和方法，例如视频-文本对比学习

2021-2023

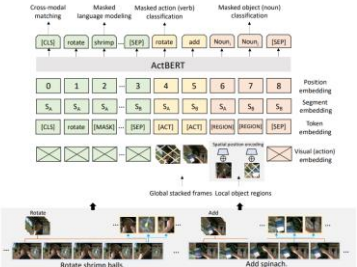
多模态集成阶段



MIL-NCE CVPR2020



VideoBERT ICCV2019

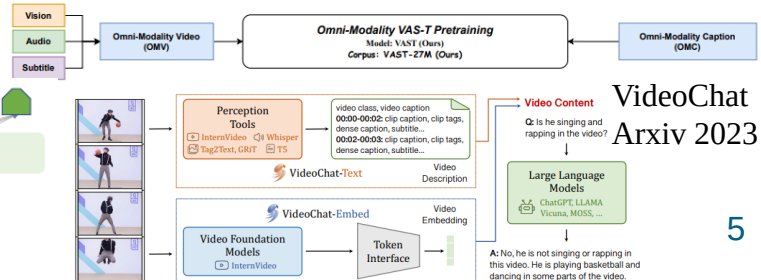
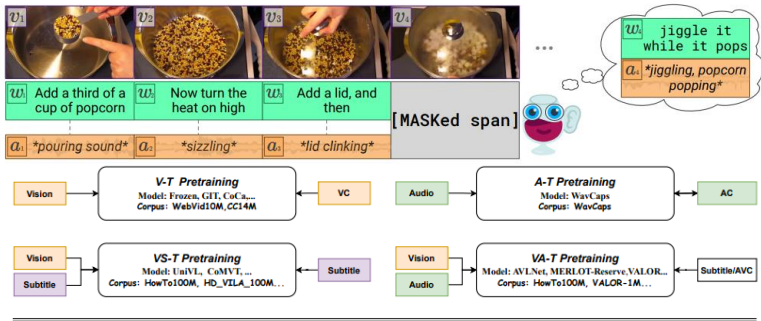
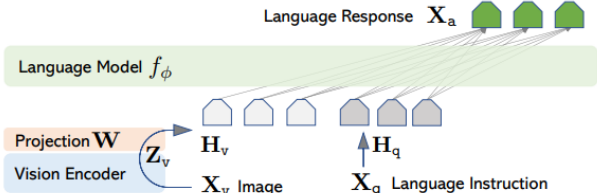


ActionBERT CVPR2020

MERLOT Reserve
CVPR 2022

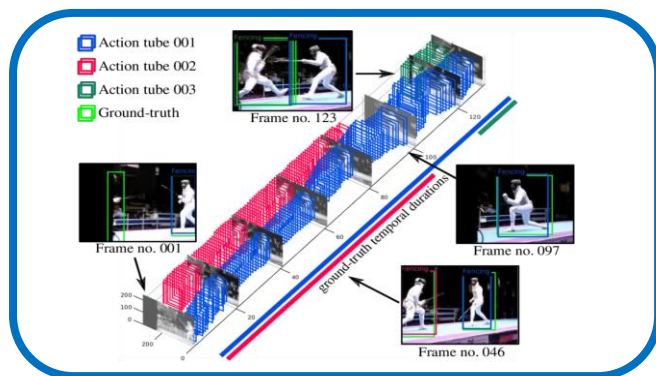
VAST
NeurIPS 2023

LLaVA
NeurIPS 2023



• 从基础动态世界感知到长时学习推理与具身交互

基础动态世界感知

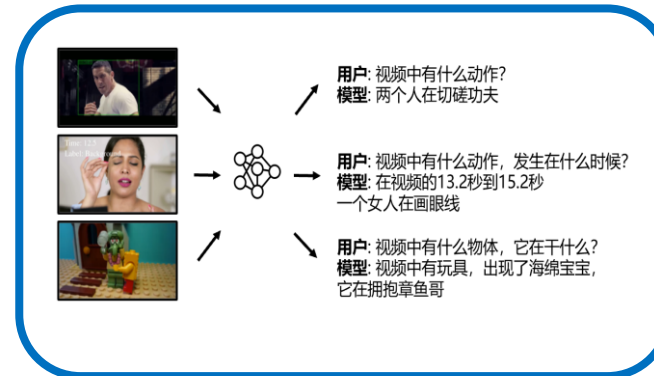


InternVideo1.0

支持了书生2.0

全球首个具备体系化动态
感知能力视频基础模型
在30余种任务SOTA

多模态开放问答理解



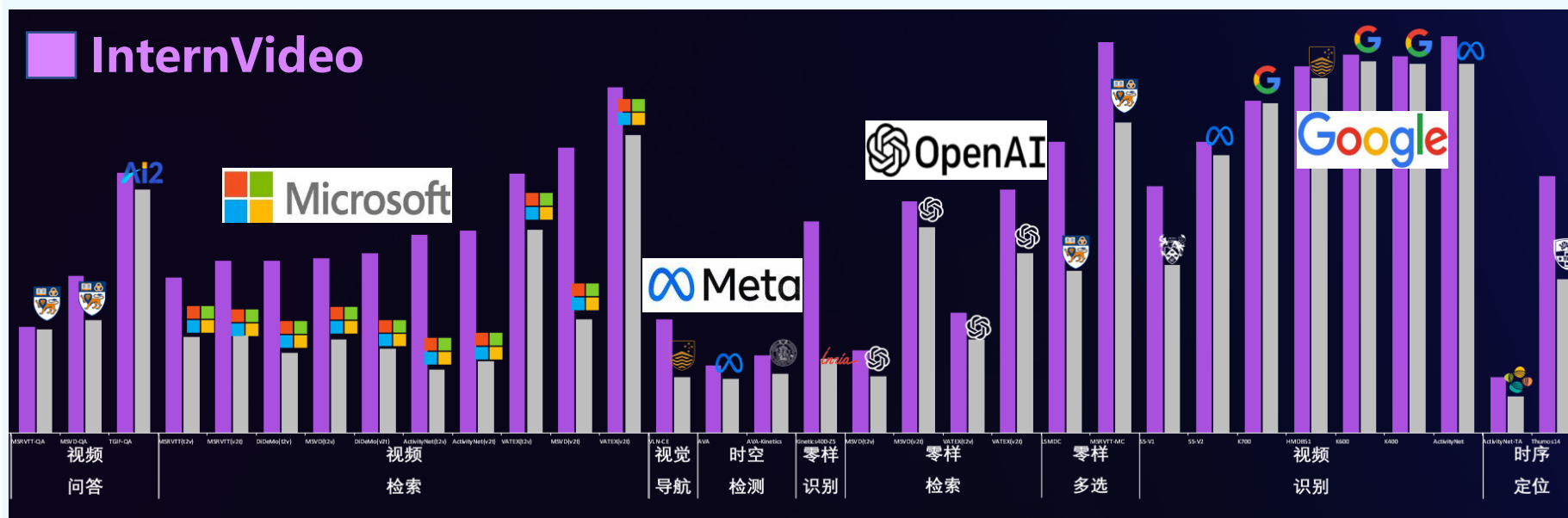
InternVideo2.0

支持了书生多模态

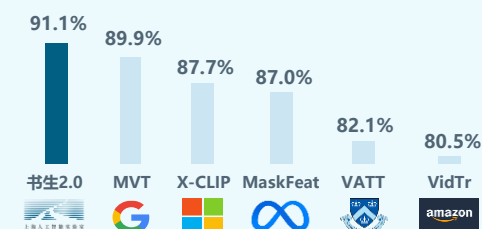
全球首个多模态
视频对话模型
在60余种任务SOTA

书生2.0-通用视频模型：InternVideo

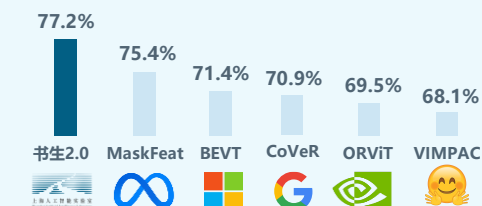
- 在视频识别、视频时序定位、视频检索等39种视频任务上精度世界领先 (截止2022年9月1日)
- 在视频识别标杆基准库Kinetics 400上首次取得超越90%的Top1准确率，达到91%



Kinetics识别准确率首次> 90%



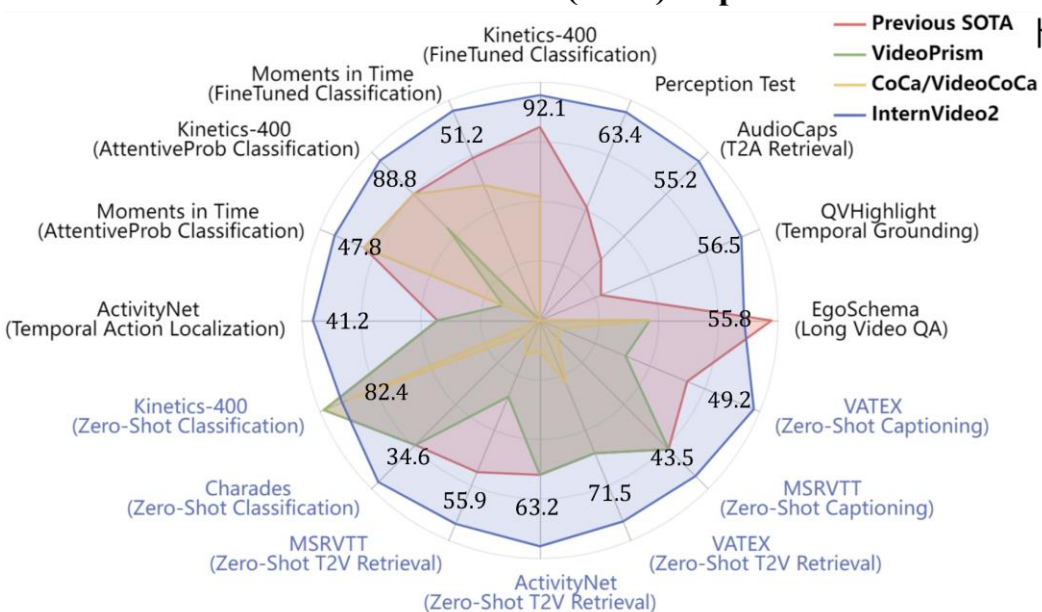
Something-Something V2



截止2022年9月1日

- 提出了一种渐进式预训练范式，逐步对齐多粒度视觉语义，形成多任务通用处理能力
- 在Scaling Law的引导下，扩大训练数据与模型规模：400M数据 & 6B视觉模型
- 在超过60种的视频理解任务上面取得了世界领先水平，涵盖多种视频理解任务

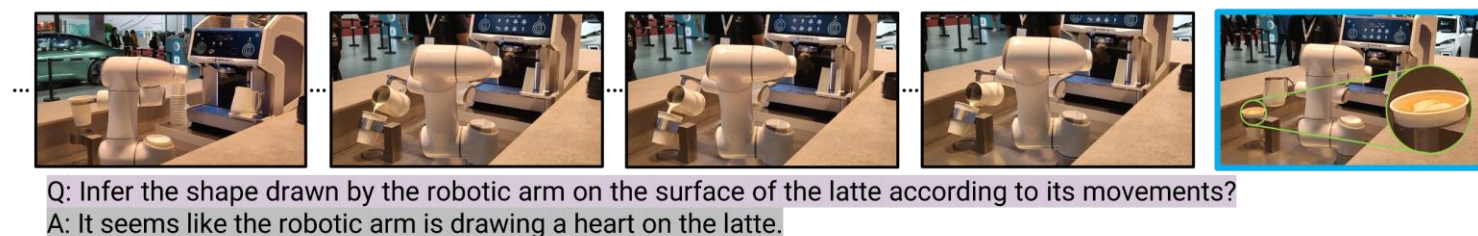
Transferable Video(-Text) Representation



Support for Long Videos

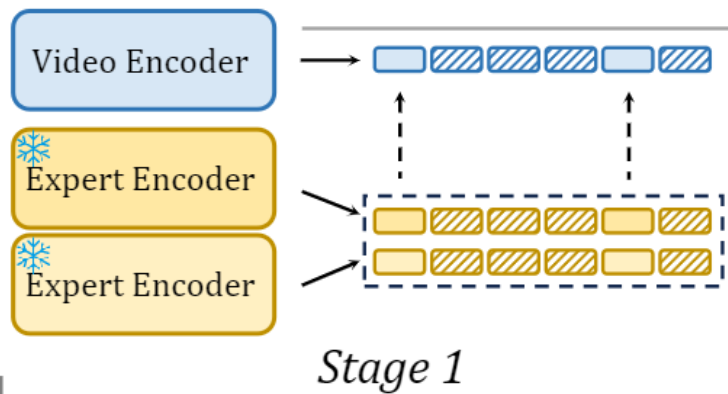


Temporal Commonsense Reasoning

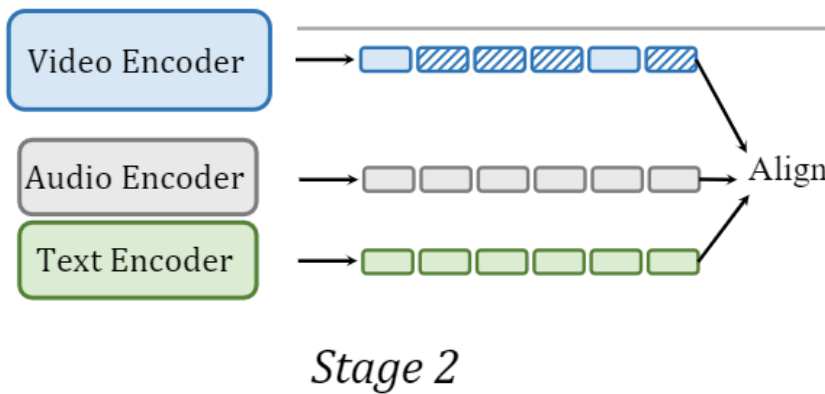


□ 渐进式预训练

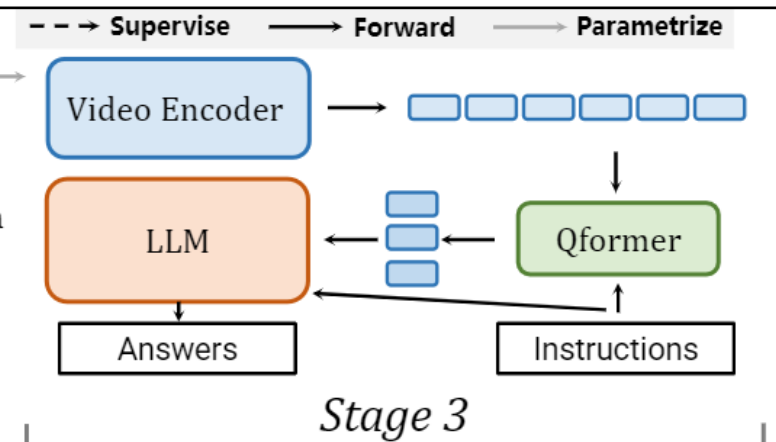
视觉特性对齐



跨模态语义对齐



多模态交互能力对齐



- 阶段1：重构未掩码视频token $\leftarrow$ InternVL-6B 1.0 + VideoMAEv2-g
- 阶段2：对齐视频表征到语音-对话-文本表征
- 阶段3：使用视频为中心的输入进行下个词元预测

□多模态视频数据

视频数据来源

Table 1: Summary of datasets used in **InternVideo2** pretraining process.

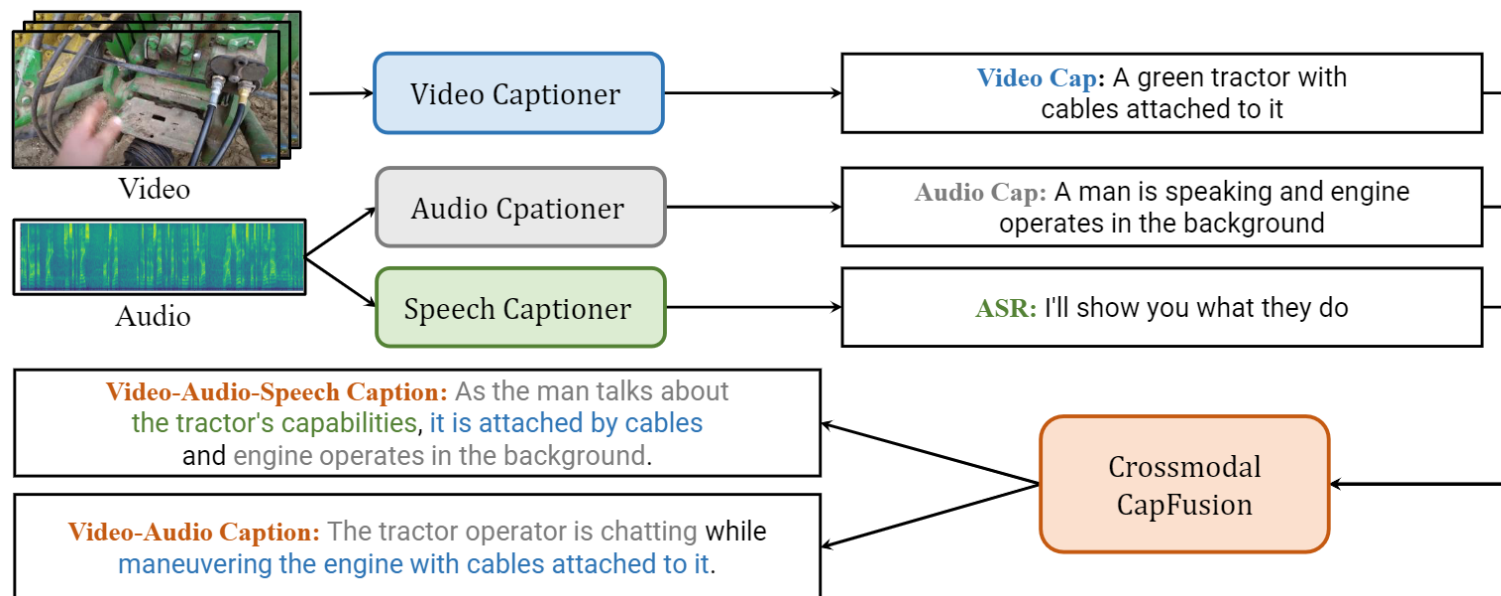
Pretraining Stage	Dataset	Domain	# of clips	Annotation
Stage 1	KMash	Web Video	2M	Action Label
Stage 2 (img-txt)	LAION, etc	Web Image	300M	Alt-text / Generated Caps
Stage 2 (vid-txt)	WebVid2M	Web Video	250k	Alt-text
	WebVid10M	Web Video	9.7M	Alt-text
	InternVid	Youtube Video	40M	Generated Caption
	Self-collected	Youtube Video	50M	Generated Caption
Stage 3	LLaVA, etc	Web Image/Video	2.1M	Conversation, QA

Table 21: Statistics of Stage1 data. All the videos are used without any labels.

Dataset	K710	SthSthV2	HACS	ANet	MiT	Self-collected
K-Mash _{1.1M}	658K	169K	106K	15K	152K	0
K-Mash _{2M}	658K	169K	106K	15K	207K	844K

□多模态视频数据

视频数据引擎



You are a text analysis expert. About one video, here is 1 vision caption: **vid_cap**, 1 audio caption: **aud_cap**. You need to understand and encode them into 1 sentence. Do not simply concatenate them together. The weights of video/audio are equal. Considering dropping audio caption if it is incomprehensible. The output must be a complete and natural sentence. The sentence is: ...

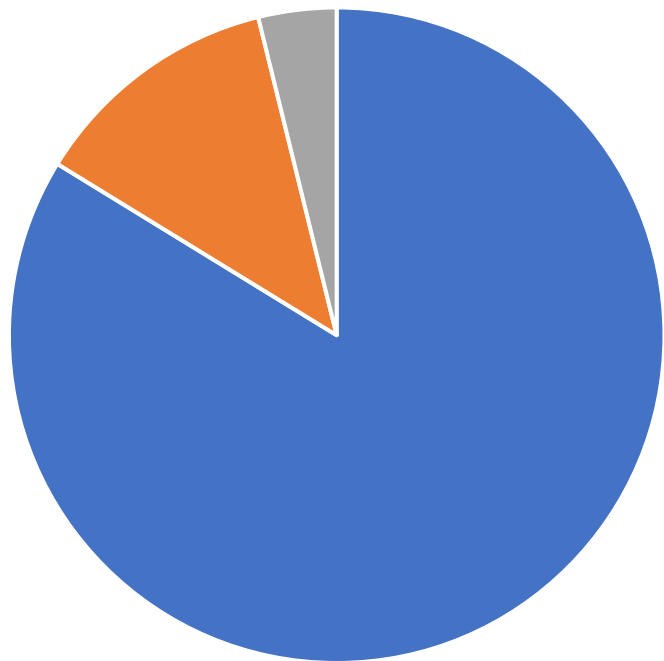
You are a text analysis expert. About one video, here is 1 vision caption: **vid_cap**, 1 audio caption: **aud_cap**, and one speech subtitle: **asr_cap**. You need to understand and encode them into 1 complete sentence. The weights of video/audio/speech are equal. Considering dropping audio caption or speech subtitle if it is incomprehensible. The output must be a complete and natural sentence, do not simply concatenate them together. The complete sentence is:

□多模态视频数据：标注实例

	Video Cap: A woman is talking to a reporter in a store. Audio Cap: A woman is talking and music plays in the background. ASR: She is gonna be our second competitor. Take a lap. She has lost 69 pounds and thirty-one percent of her body weight. Hey I'm excited but nervous but excited Carolyn.	Video-Audio-Speech Cap: In the store, music plays in the background as a woman excitedly and nervously tells a reporter about Carolyn, who is going to be their second competitor and has lost 69 pounds and 31% of her body weight. Video-Audio Cap: In a store, a woman talks to a reporter while music are heard in the background.
	Video Cap: A man in a red shirt is standing in front of a lion. Audio Cap: A man is talking while a radio plays music. ASR: Scene consisted of Denver attempting to barricade himself away from a wild lion spreading to most of the line is already in the room with him unfortunately.	Video-Audio-Speech Cap: The man in the red shirt is attempting to barricade himself away from a wild lion in a room with most of the line already there while a man is talking and the radio plays music. Video-Audio Cap: The man in the red shirt stands in front of the lion and talking while the radio plays music.
	Video Cap: A group of people are standing in a field under a starry night sky. Audio Cap: A woman is talking. ASR: When people get a chance look through a big telescope for the first time they're just floored at what you can see I was just showing the moon to someone and they're like oh my god you really can't see the craters it's really interesting.	Video-Audio-Speech Cap: A group of people are standing in a field under a starry night sky and a woman is talking in the background that a group of people observe the moon through a telescope, amazed by its craters. Video-Audio Cap: A group of people are standing in a field under a starry night sky and a woman is talking in the background.
	Video Cap: A small black fish is swimming in a pond Audio Cap: A bird is chirping while a man is talking. ASR: Come on, dude	Video-Audio-Speech Cap: While a bird is chirping in the background, a small black fish is swimming in a pond and a man is saying 'Come on, dude'. Video-Audio Cap: As a small black fish swims in a pond, a bird chirps and a man talks in the background.

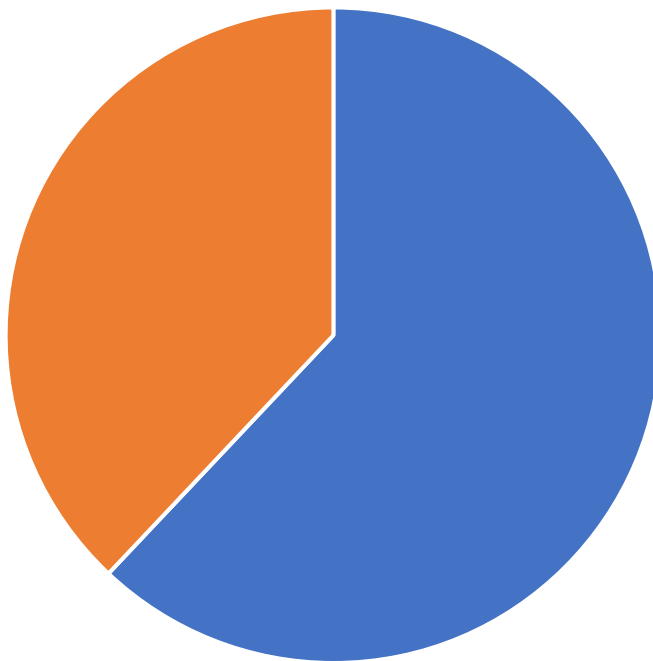
□多模态视频数据潜在偏见

年龄



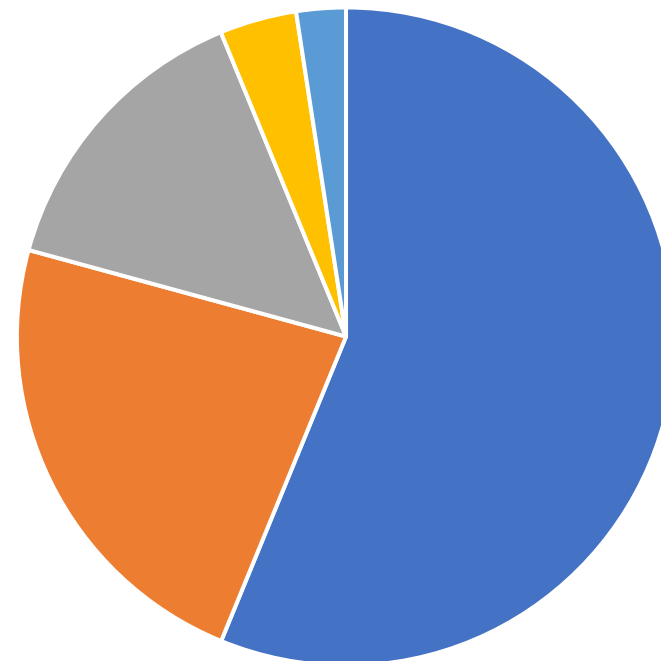
■ 成人 ■ 儿童 ■ 老人

性别



■ 男性 ■ 女性

种族



■ 亚洲人 ■ 黑人 ■ 白人
■ 中东 ■ 拉丁美洲

□ 视频演示



□动作识别：end-to-end

Table 2: End-to-end finetuning action recognition results (top-1 accuracy) on Kinetics, SomethingSomething, and Moments in Time. [†] denotes that the result is achieved with different resolutions or frame rates.

Method	Training Data	Setting	K400	K600	K700	Sth-Sthv2	MiT	ANet	HACS
CoVeR [Zhang et al., 2021]	IV-3B	16×448	87.1	87.9	79.8	70.8	46.1	-	-
Hiera-H [Ryali et al., 2023]	V-0.25M	16×224	87.8	88.8	81.1	76.5	-	-	-
CoCa-g [Yu et al., 2022]	I-3B	16×576	88.9	89.4	82.7	-	49.0	-	-
MTV-H [Wang et al., 2023b]	IV-370M	32×280	89.9	90.3	83.4	-	-	-	-
VideoMAEv2-g [Wang et al., 2023b]	V-1.35M	64×266	90.0	89.9	-	77.0 [†]	-	-	-
V-JEPA-H [Bardes et al., 2024]	V-2M	16×224	-	-	-	77.0	-	-	-
MVD-H [Wang et al., 2023c]	IV-1.25M	16×224	-	-	-	77.3	-	-	-
UniFormerV2-L [Li et al., 2022c]	IV-401M	64×336	90.0	90.1	82.7	73.0 [†]	47.8 [†]	94.7	95.4
InternVideo [Wang et al., 2022]	V-12M	<i>ensemble</i>	91.1	91.3	84.0	77.2	-	-	-
InternVideo2_{s1}-1B	IV-1.1M	8×224	91.3	91.4	85.0	77.1	50.8	-	-
InternVideo2_{s1}-1B	IV-1.1M	16×224	91.6	91.6	85.4	77.1	50.9	-	-
InternVideo2_{s1}-6B	IV-2M	8×224	91.9	91.7	85.7	77.5	51.0	-	-
InternVideo2_{s1}-6B	IV-2M	16×224	92.1	91.9	85.9	77.4	51.2	95.9	97.0

●K400/600/700 上达到 92.1%/91.9%/85.9% w/ 16帧

□动作识别：attentive probing

Table 3: Attentive probing recognition results (top-1 accuracy) on Kinetics-400/600/700, Moments in Time and Something-Something V2.

Method	Training Data	Setting	K400	K600	K700	MiT	SSV2
UMT-L [Li et al., 2023e]	IV -25M	-	82.8	-	-	40.3	54.5
VideoMAEv2-g [Wang et al., 2023b]	V -1.35M	-	82.1	-	-	35.0	56.1
V-JEPA-H [Bardes et al., 2024]	V -2M	16×384	81.9	-	-	-	72.2
DINOv2-g [Oquab et al., 2023]	I -142M	16×224	83.4	-	-	-	50.0
VideoPrism-g [Zhao et al., 2024]	V -619M	16×288	87.2	-	-	45.5	68.5
ViT-e [Dehghani et al., 2023]	I -4B	128×224	86.5	-	-	43.6	-
ViT-22B [Dehghani et al., 2023]	I -4B	128×224	88.0	-	-	44.9	-
CoCa-g [Yu et al., 2022]	I -3B	16×576	88.0	88.5	81.1	47.4	-
InternVideo2 _{s2} -1B	IV -25.5M	16×224	87.9	88.0	79.5	46.3	67.3
InternVideo2 _{s2} -6B	IV -400M	16×224	88.8	89.1	81.0	47.8	67.7

□动作识别：linear probing

Table 4: Linear probing action recognition results (top-1 accuracy) on Kinetics-400, Something-Something V2, UCF-101 and HMDB-51.

Method	Setting	K400	SSV2	UCF-101	HMDB-51
VideoMAEv2-H [Wang et al., 2023b]	12×224	25.8	-	56.4	34.1
TVTSv2-H [Zeng et al., 2023]	12×224	73.1	-	91.8	65.7
OpenCLIP-G [Cherti et al., 2023]	8×224	78.3	35.8	90.7	-
DINOv2-g [Oquab et al., 2023]	8×224	78.4	38.3	91.2	-
InternVideo2_{s1}-1B	16×224	81.6	46.3	96.0	71.6
InternVideo2_{s1}-6B	16×224	82.0	47.8	96.3	71.8
InternVideo2_{s2}-6B	16×224	84.2	56.7	97.3	80.7

□动作识别：zero-shot

Table 5: Zero-shot action recognition on UCF, HMDB, MiTv1, SSv2-MC, and Charades.

Method	#F	Training Data	UCF	HMDB	MiT	SSv2-MC	Charades
CLIP [Radford et al., 2021]	12	I-400M	68.9	43.2	-	29.6	-
TVTSv2 [Zeng et al., 2023]	12	V-8.5M	78.0	52.1	-	48.4	-
VideoCoCa-g [Yan et al., 2022]	16	V-145M	-	-	-	-	25.8
VideoPrism-g [Zhao et al., 2024]	16	V-619M	-	-	-	-	32.4
InternVideo2_{clip}-1B	8	IV-25.5M	88.8	53.9	31.6	61.5	32.9
InternVideo2_{clip}-6B	8	IV-400M	89.5	56.7	32.9	63.5	34.6

Table 6: Zero-shot action recognition results on Kinetics.

Method	#F	K400		K600		K700	
		top-1	AVG	top-1	AVG	top-1	AVG
CLIP [Radford et al., 2021]	8	58.4	70.1	55.1	67.2	46.1	58.4
EVA-CLIP-L [Sun et al., 2023b]	1	-	65.0	-	64.9	-	59.1
EVA-CLIP-E [Sun et al., 2023b]	1	-	69.8	-	69.3	-	63.4
ViCLIP-L [Wang et al., 2023d]	8	64.8	75.7	62.2	73.5	54.3	66.4
VideoCoCa-g [Yan et al., 2022]	16	72.0	81.3	-	-	-	-
InternVL-6B [Chen et al., 2023c]	8	69.1	79.4	68.9	78.8	60.6	71.5
EVA-CLIP-18B [Sun et al., 2024]	16	-	79.4	-	79.4	-	72.2
VideoPrism-g [Zhao et al., 2024]	16	76.4	85.4	-	-	-	-
InternVideo2_{clip}-1B	8	<u>73.1</u>	<u>82.4</u>	72.8	81.8	64.9	75.2
InternVideo2_{clip}-6B	8	72.7	82.2	71.7	81.2	64.2	75.2

□时序动作识别

Table 7: Finetuned temporal action localization results on THUMOS14 [Idrees et al., 2017], ActivityNet [Krishna et al., 2017], HACS Segment [Zhao et al., 2019] and FineAction [Liu et al., 2022]. We report average mAP. “Flow” denotes the ensembling I3D flow feature. * denotes the result is achieved with Flow.

Backbone	THUMOS14	HACS	ActivityNet	FineAction
I3D [Carreira and Zisserman, 2017] + Flow	66.8	-	35.6	-
R(2+1)D [Tran et al., 2018]	55.6	-	36.6	-
InternVideo	71.6*	41.3	39.0	17.6
VideoMAEv2-g [Wang et al., 2023b]	69.5	-	-	18.2
InternVideo2_{s1}-1B	69.8	42.4	40.4	27.2
InternVideo2_{s1}-6B	72.0	43.3	41.2	27.7

□时序定位

(a) QVHighlight

Feature	R1@0.5	R1@0.7	mAP	mAP	HiT@1
CLIP [Radford et al., 2021]	64.97	48.65	42.96	39.83	64.19
CLIP+SlowFast [Feichtenhofer et al., 2019]	65.43	48.38	42.86	40.33	66.21
InternVideo2_{s2}-1B	<u>70.00</u>	<u>54.45</u>	<u>47.02</u>	<u>42.36</u>	<u>69.74</u>
InternVideo2_{s2}-6B	71.42	56.45	49.24	42.90	72.00

(b) Charade-STA

Feature	R1@0.3	R1@0.5	R1@0.7	mIoU
CLIP [Radford et al., 2021]	65.62	52.77	30.16	45.85
CLIP+SlowFast [Feichtenhofer et al., 2019]	70.43	58.44	36.34	50.13
InternVideo2_{s2}-1B	<u>78.41</u>	<u>68.36</u>	<u>45.03</u>	<u>57.12</u>
InternVideo2_{s2}-6B	79.70	70.03	48.95	58.79

□视频实例分割

Table 8: Video instance segmentation performance (mAP) on YouTube-VIS19 [Yang et al., 2019].

Method	Backbone	#Params	YouTubeVIS19
Mask2Former	Swin-L (image) [Liu et al., 2021]	219M	60.3
Mask2Former	InternViT (image)	6B	63.4
Mask2Former	InternVideo2_{s1}	6B	64.2

□ 视频检索

Table 9: Results of zero-shot video retrieval in both text-to-video (T2V) and video-to-text (V2T) on MSR-VTT, LSMDC, DiDeMo, MSVD, ActivityNet (ANet), and VATEX.

Method	MSR-VTT		LSMDC		DiDeMo		MSVD		ANet		VATEX	
	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T
CLIP [Radford et al., 2021]	30.4	24.2	13.9	11.9	12.7	18.7	40.5	57.2	9.1	13.2	-	-
CLIP4Clip [Luo et al., 2022]	32.0	-	15.1	-	-	-	38.5	-	-	-	-	-
ViCLIP [Wang et al., 2023d]	42.4	41.3	20.1	16.9	18.4	27.9	49.1	75.1	15.1	24.0	-	-
InternVideo-L [Wang et al., 2022]	40.7	39.6	17.6	13.2	31.5	33.5	43.4	67.6	30.7	31.4	49.5	69.5
UMT-L [Li et al., 2023e]	40.7	37.1	24.9	21.9	48.6	49.9	49.0	74.5	41.9	39.4	-	-
VideoCoCa-g [Yan et al., 2022]	34.4	64.7	-	-	-	-	-	-	34.5	33.0	53.2	73.6
VideoPrism-g [Zhao et al., 2024]	39.7	71.0	-	-	-	-	-	-	52.7	50.3	62.5	77.1
InternVideo2_{s2}-1B	51.9	50.9	32.0	27.3	57.0	54.3	58.1	83.3	60.4	54.8	70.4	85.4
InternVideo2_{s2}-6B	55.9	53.7	33.8	30.1	57.9	57.1	59.3	83.1	63.2	56.5	71.5	85.3

Table 10: Results of finetuning video retrieval in both text-to-video (T2V) and video-to-text (V2T) on MSR-VTT, LSMDC, DiDeMo, MSVD, ActivityNet (ANet), and VATEX.

Method	MSR-VTT		LSMDC		DiDeMo		MSVD		ANet		VATEX	
	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T
CLIP [Radford et al., 2021]	38.2	38.7	22.5	22.6	32.2	33.9	-	-	26.1	26.9	-	-
CLIP4Clip [Luo et al., 2022]	45.6	45.9	24.3	23.8	43.0	43.6	45.2	48.4	40.3	41.6	-	-
ViCLIP [Wang et al., 2023d]	52.5	51.8	33.0	32.5	49.4	50.2	-	-	49.8	48.1	-	-
UMT-L [Li et al., 2023e]	58.8	58.6	43.0	41.4	70.4	65.7	58.2	82.4	66.8	64.4	72.0	86.0
InternVideo2_{s2}-6B	62.8	60.2	46.4	46.7	74.2	71.9	61.4	85.2	74.1	69.7	75.5	89.3

语音检索和问答

Table 12: Audio retrieval results on AudioCaps [Kim et al., 2019], Clothov1, and Clothov2 [Drossos et al., 2020]. We report text-to-audio R@1 accuracy in zero-shot and finetuning settings.

Method	Zero-shot			Finetuning		
	AudioCaps	ClothoV1	ClothoV2	AudioCaps	ClothoV1	ClothoV2
VIP-ANT [Zhao et al., 2021]	27.7	-	-	-	-	-
VAST [Chen et al., 2024b]	-	-	-	52.0	25.1	26.9
LanguageBind [Zhu et al., 2023a]	-	12.1	12.1	-	-	-
InternVideo2_{s2}-6B	37.1	17.4	17.4	55.2	25.3	27.2

(a) ClothoAQA and AMAVQA

Backbone	ClothQA	AMAVQA
AquaNet [Lipping et al., 2022]	14.78	65.59
MWAFM [Li et al., 2023b]	22.24	67.54
InternVideo2_{s2}	30.14	80.51

(b) ESC-50

Method	Top-1 Acc
AST [Gong et al., 2021]	95.60
BEATs [Chen et al., 2023a]	98.10
InternVideo2_{s2}	98.60

□ 视频对话

Table 14: Results of Chat-centric Evaluation on Multiple Choice Video-QA on MVBench [Li et al., 2023d], Egoschema [Mangalam et al., 2023] and Perception Test [Patraucean et al., 2024].

Model	ViEncoder	LLM	MVBench	Egoschema	Perception Test
GPT-4V [OpenAI, 2023b]	-	GPT-4	43.5	-	-
Gemini 1.0 Pro [Team et al., 2023]	-	-	37.7	55.7	51.1
Gemini 1.0 Ultra [Team et al., 2023]	-	-	-	61.5	54.7
Gemini 1.5 Pro [Team et al., 2023]	-	-	-	72.2	-
LLaVA-Next-Video [Liu et al., 2024]	CLIP-L	Vicuna-7B	46.5	43.9	48.8
VideoLLaMA2 [Cheng et al., 2024]	CLIP-L-336	Mistral-7B	54.6	51.7	51.4
VideoLLaMA2 [Cheng et al., 2024]	CLIP-L-336	Mistral-8*7B	53.9	53.3	52.2
VideoChat2	UMT-L	Vicuna-7B	51.1	-	-
VideoChat2	InternVideo2_{s3}-1B	Mistral-7B	60.3	55.8	53.0
VideoChat2-HD	InternVideo2_{s3}-1B	Mistral-7B	65.4	60.2	60.1
VideoChat2-HD-F16	InternVideo2_{s3}-1B	Mistral-7B	67.2	60.0	63.4

□消融：放缩视频编码器

Table 15: Average top-1 accuracy of action recognition (K400, SSv2, and MiT) and video retrieval (MSR-VTT, LSMDC, DiDeMo, MSVD, ANet, and VATEX in t2v) using zero-shot and finetuning settings. * denotes results are from **InternVideo2_{s1}**.

Model	Zero-shot		Finetuning
	Action Recognition	Video Retrieval	Action Recognition
InternVideo2_{s2}-1B	55.5	55.0	73.2*
InternVideo2_{s2}-6B	56.9 _(+1.4)	56.9 _(+1.9)	73.6 _(+0.4) *

□消融：一阶段训练数据和专家模型

Table 16: Ablation of Stage1, conducted with finetuned action recognition on Kinetics, MiT, and SthSthv2. All models are tested with $8 \times 224 \times 224$ input.

Model	Teacher	Data	K400	K600	K700	MiT	SSv2	Avg
ViT-L	CLIP-L	K710	90.3	90.4	83.2	48.0	74.7	77.3
ViT-L	CLIP-L	K-Mash _{1.1M}	90.5	90.4	83.4	48.1	74.7	77.4
ViT-1B	InternVL-6B	K710	90.9	91.0	84.7	49.8	75.9	78.5
ViT-1B	InternVL-6B	K-Mash _{1.1M}	91.4	91.5	85.1	50.5	76.5	79.0
ViT-1B	InternVL-6B+VideoMAE-g	K-Mash _{1.1M}	91.3	91.4	85.0	50.8	77.1	79.1
ViT-1B	InternVL-6B	K-Mash _{2M}	91.3	91.5	85.1	50.6	76.6	79.0
ViT-6B	InternVL-6B+VideoMAE-g	K-Mash _{2M}	91.9	91.7	85.7	51.0	77.5	79.6

□消融：二阶段训练方法和数据

Table 17: Ablation of Stage2. All models are tested with $8 \times 224 \times 224$ input.

Method	MSR-VTT
Baseline (video & text encoders w/ video-text learning)	24.7
Baseline + audio encoder w/ audio-text learning	24.0
Baseline + speech encoder w/ video-speech-text learning	24.9
Baseline + audio encoder w/ video-audio-text learning	27.8
Baseline + audio & speech encoders + video-audio-speech-text learning	25.7

Table 18: Zero-shot t2v retrieval on MSR-VTT with different training captions.

SceneDet	AutoShot	Video Cap	Audio Cap	Speech Cap	MSR-VTT
✓		✓			24.7
✓		✓	✓		26.6
✓		✓	✓	✓	27.1
	✓	✓	✓	✓	34.8

□消融：三阶段训练和评估

Table 19: Ablation on using qformer instruction in Stage3 training of Chat-Centric Model.

Model	MVBench	NextQA	Egoschema-full	Egoschema-subset
VideoChat2 w qformer inst	59.9	79	52.9	65.8
VideoChat2 w/o qformer inst	60.4 (+0.5)	78.6 (-0.4)	55.8 (+2.9)	66.4 (+0.6)

- 向QFormer添加问题会导致某种程度的过拟合

未来工作：长时学习推理与决策执行

WORLD MODEL ON MILLION-LENGTH VIDEO AND LANGUAGE WITH BLOCKWISE RINGATTENTION

Hao Liu*, Wilson Yan*, Matei Zaharia, Pieter Abbeel
UC Berkeley

Text: Books3 Doc Length 200k - 500k Context: 256k Tokens: 10B Examples: 37K	Video (100 frames): Valley-Instruct-73k + Video-Instruct 100k  Context: 32k Tokens: 3.2B Examples: 173K User: What is happening in the video? Assistant: A woman is dealing cards and chips at a black jack table... t = 0s t = 24s
Text: Books3 Doc Length 500k - 1M Context: 512k Tokens: 3B Examples: 3.5K	Video (450 frames): Valley-Instruct-73k + Video-Instruct 100k  Context: 128k Tokens: 1.2B Examples: 173K User: What are some of the highlights of the video? Assistant: The video captures various exhilarating moments of dirt biking, including bikers jumping into the air, ... t = 0s t = 2min
Text: Books3 Doc Length: 1M+ Context: 1M Tokens: 1B Examples: 0.8K	Video (4000 frames): Valley-Instruct-73k + Video-Instruct 100k  Context: 1M Tokens: 200M Examples: 173K User: Could you provide a brief summary of the employee's actions? Assistant: In the video, an employee prepares a sub. After assembling the bread, ham, pepperoni, salami, and cheese, he toasts the sub in the oven... t = 0s t = 10min

Towards Generalist Robot Learning from Internet Video: A Survey

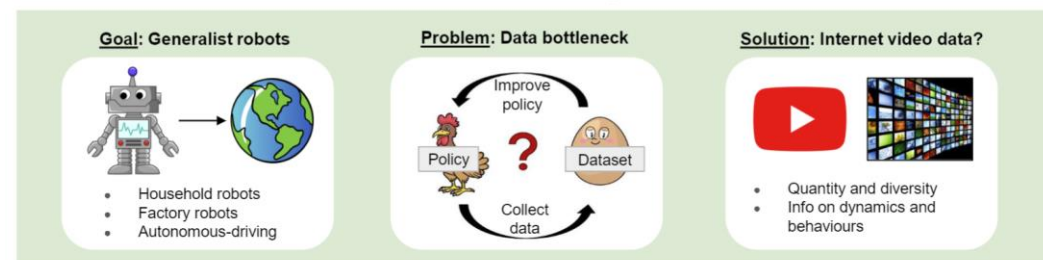
Robert McCarthy¹, Daniel C.H. Tan¹, Dominik Schmidt², Fernando Acero¹,
Nathan Herr¹, Yilun Du³, Thomas G. Thuruthel¹, and Zhibin Li¹

¹University College London

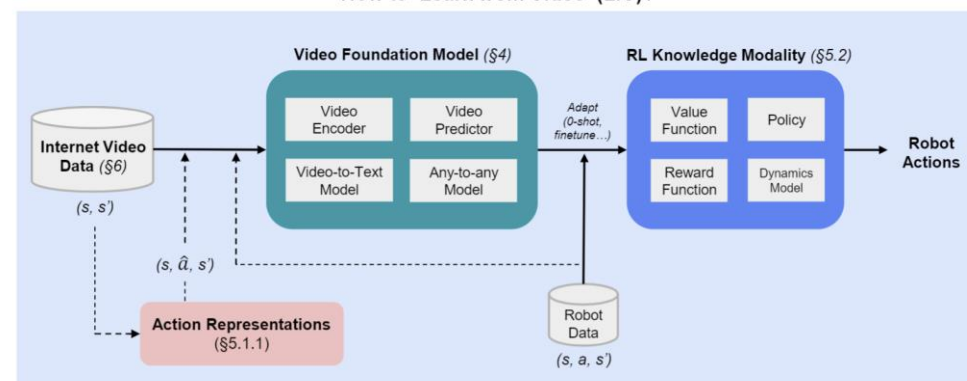
²Weco AI

³Massachusetts Institute of Technology

Towards Generalist Robot Learning from Internet Video



How to 'Learn from Video' (LfV)?



以长视频数据为中心构建动态世界基础模型，形成长时记忆、关联、推理、预测、规划、决策等能力



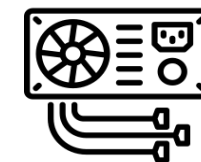
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专业协作



大牛指导



充沛算力

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Models 84		Sort: Recently updated
OpenGVLab/InternVideo2-Chat-8B	Image-Text-to-Text • Updated about 19 hours ago • 3.65k • 8	OpenGVLab/InternVideo2_chat_8B_HD
OpenGVLab/videochat	Updated about 20 hours ago	OpenGVLab/ShareGPT-4o
OpenGVLab/InternVideoMAE_models	Updated 2 days ago	OpenGVLab/InternVideo2_distillation_models
OpenGVLab/InternOmni	Updated 3 days ago • 13	OpenGVLab/VBench_Used_Models
OpenGVLab/InternVL2-2B-AWQ	Image-Text-to-Text • Updated 5 days ago • 646 • 13	OpenGVLab/InternVL2-40B

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谢 谢!